

Vietnam National University

Ho Chi Minh University of Science

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FINAL PROJECT REPORT

Topic: SINO-NÔM TO ROMANIZED VIETNAMESE TRANSLITERATION

Subject: Statistical Machine Learning

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1 Introduction

The Sino-Nôm script is an ideographic, logographic writing system developed by the Vietnamese people based on Chinese characters to transcribe the spoken Vietnamese language. For more than ten centuries (from the 10th to the 20th century), Sino-Nôm script, alongside Classical Chinese (Han), served as the primary and official medium for recording and preserving the cultural heritage, history, literature, and administrative documents of the Vietnamese nation. Classical masterpieces such as *Truyện Kiều* (by Nguyễn Du) and *Quốc âm thi tập* (by Nguyễn Trãi), as well as royal decrees and genealogical records, were all documented in this writing system.

However, with the complete replacement by modern Vietnamese (the Latin-based script) in the modern era, Sino-Nom has become largely unfamiliar and faded into obscurity for the vast majority of Vietnamese people. Nevertheless, because an immense repository of Vietnam’s historical documents and intellectual legacy remains recorded in Sino-Nom, the preservation, research, and systematization of this writing system are of vital importance. Today, leveraging computer science to develop computational tools for translation, transliteration, and text structuring between Sino-Nom and modern Vietnamese offers substantial practical value toward these objectives. Digitizing and constructing automated conversion tools between modern Sino-Nom and Vietnamese not only hold profound significance for cultural heritage preservation, but also present an intriguing and challenging task in the field of Natural Language Processing (NLP).

This project aims to build a web-based transliteration application from Sino-Nôm to modern Vietnamese, using curated dataset from historical literature and Neural Machine Translation. It also analyzes and identifies the nuances and differences between 2 models that were used for this task: mBART and mT5.

2 Problem Definition

2.1 Transliteration and Linguistic Ambiguity

Transliteration is defined as the process of converting text from one writing system to another within the same language or across linguistic boundaries [1]. Common examples of this conversion include mapping from the Cyrillic script to the Latin script (e.g., “Владимир Путин” → “Vladimir Putin”), or from the Latin script (Romaji) to Katakana (e.g., “Mike Tyson” → “マイク・タイソン”). This character-level conversion fundamentally differs from translation, as transliteration focuses primarily on phonetic

representation and orthography rather than transferring semantic meaning. In alphabetic or syllabic scripts, transliteration can often be accomplished straightforwardly via direct character-to-character or character-to-phoneme mapping.[1]

However, such straightforward mapping is feasible because those scripts are phonographic (phonetic), where graphemes directly reflect spoken pronunciation. In sharp contrast, Sino-Nom is an ideographic and logographic writing system, wherein each character encapsulates a specific semantic and morphemic unit. Consequently, transliterating from Sino-Nom to Romanized Vietnamese introduces severe one-to-many ambiguities (polyphony and heteronymy), where a single Sino-Nom character frequently corresponds to multiple valid Romanized Vietnamese syllables.

For example:

- The character 沒 can be read as “một” (meaning the numeral one) or “mất” (meaning to lose or pass away).
- The character 行 can be read as “hành” (as in *hành động* - to act) or “hàng” (as in *hàng ngũ* - rank/row).
- The character 共 can be read with its Sino-Vietnamese sound “cộng” (together/total) or its vernacular Nom sound “cùng” (with/and).

Hence, accurate transliteration cannot rely on static dictionary lookups; it necessitates dynamic disambiguation based on surrounding context, syntactic structures, and historical-cultural knowledge.

2.2 Theoretical Background

2.2.1 Neural Machine Translation

Machine Translation (MT) is a vital subfield of Artificial Intelligence (AI) and Natural Language Processing (NLP) that investigates computational methods for automatically converting text or speech from a source language into a target language.

Mathematically, the objective of a machine translation system is to find the optimal target sequence $\hat{Y}$ that maximizes the conditional probability given a source sequence X :

$$\hat{Y} = \arg \max_Y P(Y | X) \quad (1)$$

Within the scope of this project, “language” is interpreted in a broad sense. The task of transliterating from Sino-Nom to Romanized Vietnamese is formulated as a monolingual machine translation or script conversion task, where the input $X = (x_1, x_2, \dots, x_n)$ represents a sequence of ideographic Sino-Nom characters, and the output $Y = (y_1, y_2, \dots, y_m)$ represents the corresponding sequence of Romanized Vietnamese syllables.

Due to the pervasive polyphonic ambiguities inherent in Sino-Nom, conventional paradigms such as Rule-Based Machine Translation (RBMT) or Statistical Machine Translation (SMT) exhibit severe limitations, as they struggle to capture long-range semantic context. Therefore, Neural Machine Translation (NMT)—which relies on deep artificial neural networks—is the most suitable framework for this task. Mathematically, NMT models the conditional probability of the target sequence Y conditioned on the source sequence X through the chain rule of probability:

$$P(Y | X) = \prod_{t=1}^m P(y_t | y_{<t}, X; \theta) \quad (2)$$

where θ denotes the set of trainable parameters of the model, and $y_{<t} = (y_1, \dots, y_{t-1})$ represents the sequence of previously generated target tokens.

The canonical architecture for NMT is the **Encoder-Decoder** framework:

- **Encoder:** Transforms the variable-length input sequence X into continuous latent semantic vector representations.
- **Decoder:** Autoregressively utilizes these latent representations to generate the target sequence Y token by token.

For Sino-Nom transliteration, NMT is exceptionally effective at resolving polyphonic ambiguities by utilizing contextual embeddings across the entire sequence.

2.2.2 The Transformer Architecture

Originating from the seminal paper titled “Attention Is All You Need” [5], the Transformer is widely regarded as the most monumental breakthrough in modern NLP history, completely supplanting the long-standing dominance of Recurrent Neural Networks (RNN/LSTM) in sequence processing tasks.

Unlike RNNs, which process sequential data autoregressively (one word at a time), the Transformer relies entirely on the Attention mechanism to model dependencies between tokens regardless of their

distance in the sentence, thereby enabling full parallelization across the entire sequence. In particular, the Transformer introduces the **Self-Attention** mechanism, which empowers the model to resolve lexical ambiguities by leveraging contextual information. This mechanism simulates an information retrieval process based on three vectors:

- **Query (Q):** Represents the current token under consideration.
- **Key (K):** Represents other tokens in the sentence (used for comparison).
- **Value (V):** Encapsulates the actual semantic content of those tokens.

The computation process is carried out in the following steps:

1. Compute the similarity scores between the current token (Q) and all other tokens (K) via the dot product.
2. Normalize the resulting scores using the Softmax function to obtain the **Attention Weights**.
3. Multiply these weights with V to aggregate the most relevant contextual information.

Mathematically, the Scaled Dot-Product Attention is formulated as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V \quad (3)$$

where d_k denotes the dimension of the Key vectors.

The Transformer architecture serves as the primary backbone for all models deployed in this project.

2.3 Approaching the problem

Based on the theoretical foundations established above, this project frames the transliteration from Sino-Nom to Romanized Vietnamese as an end-to-end Neural Machine Translation problem. To overcome the extreme low-resource constraint of historical Sino-Nom texts, we adopt a transfer learning strategy by leveraging two state-of-the-art multilingual Transformer models: **mT5** and **mBART**.

2.3.1 Model Architectures and Selection Rationale

- **mT5 (Multilingual Text-to-Text Transfer Transformer):** mT5 is an encoder-decoder Transformer architecture that frames all NLP tasks into a unified text-to-text format. Unlike standard Transformers that use absolute positional encodings, mT5 employs relative position biases, allowing it to generalize more effectively across varying sequence lengths. mT5 was pre-trained on the massive mC4 corpus covering 101 languages using a self-supervised span-corruption (masked language modeling) objective, where arbitrary spans of input tokens are masked with sentinel tokens and the decoder learns to reconstruct them.
- **mBART (Multilingual Denoising Pre-trained Sequence-to-Sequence):** mBART is a sequence-to-sequence denoising autoencoder based on the standard BART architecture, consisting of a bidirectional encoder coupled with an autoregressive decoder. It was pre-trained by applying full-text corruption (such as arbitrary text span masking, token deletion, and sentence permutation) across large-scale monolingual corpora in up to 50 languages (CC50), training the model to reconstruct the original clean text.

Rationale for Model Selection in Sino-Nom Transliteration: The fundamental motivation for selecting mT5 and mBART lies in their extensive multilingual pre-training coverage, which explicitly includes both Chinese and Vietnamese:

1. **Visual and Structural Knowledge from Chinese:** Sino-Nom characters share a vast majority of their graphic components, radicals, and ideological structures with Chinese Hanzi. Pre-training on massive Chinese corpora equips these models with rich prior representations of logographic character topologies and semantic radicals.
2. **Linguistic and Syntactic Knowledge from Vietnamese:** Both models were pre-trained on modern Vietnamese corpora, enabling them to acquire strong syntactic understanding, rich vocabulary, and the intricate tonal and diacritical system of Romanized Vietnamese.
3. **Cross-lingual Transfer for Low-resource Data:** Historical Sino-Nom is a critically low-resource script with limited digitized parallel datasets. Training a neural model from scratch on such scarce data typically leads to severe overfitting. By fine-tuning mT5 and mBART, the models can effectively transfer their existing knowledge of Chinese character structures to decode Sino-Nom characters and map them into fluent, contextually accurate Romanized Vietnamese sentences.

2.3.2 Evaluation Metrics

To rigorously quantify the quality of the generated Romanized Vietnamese transcriptions, we employ a comprehensive set of evaluation metrics spanning precision, recall, and character-level edit distance:

- **BLEU (Bilingual Evaluation Understudy)**: Evaluates the precision of n -gram overlaps between the generated sequence and the ground-truth references, reflecting the fluency and lexical accuracy of the transliteration.
- **ROUGE (Recall-Oriented Understudy for Gisting Evaluation)**: Evaluates the recall capability of the model. In particular, **ROUGE-L** measures the Longest Common Subsequence (LCS) between the generated output and reference text, assessing whether the overall sentence structure and word order are faithfully preserved.
- **WER (Word Error Rate)**: Measures the minimum edit distance at the word and syllable level, calculated as:

$$\text{WER} = \frac{S + D + I}{N} \quad (4)$$

where S , D , and I denote substitutions, deletions, and insertions, respectively, and N is the total number of words in the ground truth. WER provides an intuitive assessment of how many syllable-level corrections a human editor would need to perform.

2.3.3 Application Deployment

Following rigorous training, hyperparameter tuning, and metric evaluation, the best-performing model is packaged and deployed into an interactive, user-friendly **web-based application**. This system allows end-users, historians, and researchers to input raw Sino-Nom text and automatically obtain accurate, context-aware Romanized Vietnamese transliterations, thereby providing a practical computational tool for cultural heritage preservation.

3 Dataset and Exploratory Data Analysis (EDA)

3.1 Dataset Provenance and Literary Origins

3.1.1 Direct Data Source and Curation

The empirical dataset utilized in this research is sourced and curated from the open-access **NomNa OCR** project, hosted on the Kaggle repository [?]¹. The NomNa OCR initiative is a specialized historical text-digitization effort designed to transcribe scanned historical Sino-Nom archives into machine-readable characters paired synchronously with standardized modern Romanized Vietnamese (Chữ Quốc ngữ). The primary material consists of 2,956 **high-resolution scanned pages** of authentic Vietnamese historical woodblock prints (*bản khắc gỗ*) and hand-copied manuscripts. Through optical character recognition, page layout segmentation, and meticulous philological cross-referencing by Sino-Nom scholars, each page was transcribed into discrete, paired lines of text, yielding an initial raw corpus of 30,415 **parallel sentence pairs**.

3.1.2 Historical and Philological Origins

The underlying texts represent the foundational pillars of Vietnamese historiography and classical literature, spanning distinct literary genres, centuries, and regional dialects:

1. **Đại Việt Sử Ký Toàn Thư (The Complete Annals of Dai Viet - 大越史記全書)**: Compiled under the Later Lê Dynasty by court historian Ngô Sĩ Liên and subsequent imperial chroniclers, this is the paramount official historical record of feudal Vietnam. Written entirely in Classical Chinese (Chữ Hán), the corpus encompasses five major historical volumes comprising 2,453 **pages** (68.11% of the dataset):
 - *Quyển Thủ* (Introductory Volume): 107 pages.
 - *Ngoại ký toàn thư* (Peripheral Annals): 178 pages.
 - *Bản kỷ toàn thư* (Basic Annals Complete Book): 933 pages.
 - *Bản kỷ thực lục* (Veritable Records): 787 pages.
 - *Bản kỷ tục biên* (Continued Basic Annals): 448 pages.

¹<https://www.kaggle.com/datasets/quandang/nomnaocr>

This source provides a rigorous linguistic foundation characterized by formal historiographical prose, bureaucratic terminology, and thousands of historical named entities.

2. **Truyện Kiều (The Tale of Kieu - 斷腸新聲)**: Authored by the national poet Nguyễn Du in the early 19th century, this masterpiece represents the zenith of Vietnamese vernacular poetry written in Chữ Nôm. To capture authentic orthographic diversity and historical variations in Nôm glyph structures, the dataset incorporates three distinct historical woodblock editions comprising 399 pages (25.18% of the corpus):

- *Bản Liễu Văn Đường (1866)*: 100 pages.
- *Bản 1871*: 136 pages.
- *Bản Duy Minh Thị (1872)*: 163 pages.

Each edition provides valuable variations in Nôm orthography set within the strict rhythmic structure of the classical *lục bát* meter.

3. **Lục Vân Tiên (蓼雲仙)**: Composed in the mid-19th century by the celebrated southern poet Nguyễn Đình Chiểu, this epic narrative poem spans 104 pages (2,040 lines, representing 6.71% of the corpus). Written in vernacular Chữ Nôm, it contributes essential linguistic diversity to the dataset by incorporating authentic 19th-century Southern Vietnamese colloquialisms and phonetic variants.

3.2 Fundamental Nature: Transliteration versus Translation

A critical prerequisite in formulating our computational pipeline is distinguishing between translation and transliteration:

- **Machine Translation (Semantic Transfer)**: Typically exhibits non-linear syntactic reordering and non-monotonic word alignments, resulting in substantial divergence between source and target sentence lengths.
- **Transliteration (Script Conversion)**: Maps ideographic characters (Sino-Nom) directly to phonographic syllables (Romanized Vietnamese). This task exhibits a strictly **monotonic property** with a near-perfect 1 : 1 **mapping ratio** (1 Sino-Nom character $\approx$ 1 Romanized Vietnamese syllable).

Substantial deviations from this ratio invariably indicate textual corruption, OCR segmentation artifacts, or translation paraphrasing in the raw corpus.

3.3 Corpus Composition and Source Distribution

Table 1 provides a granular breakdown of the raw corpus files, including their genre classification, line volumes, and mean sequence lengths.

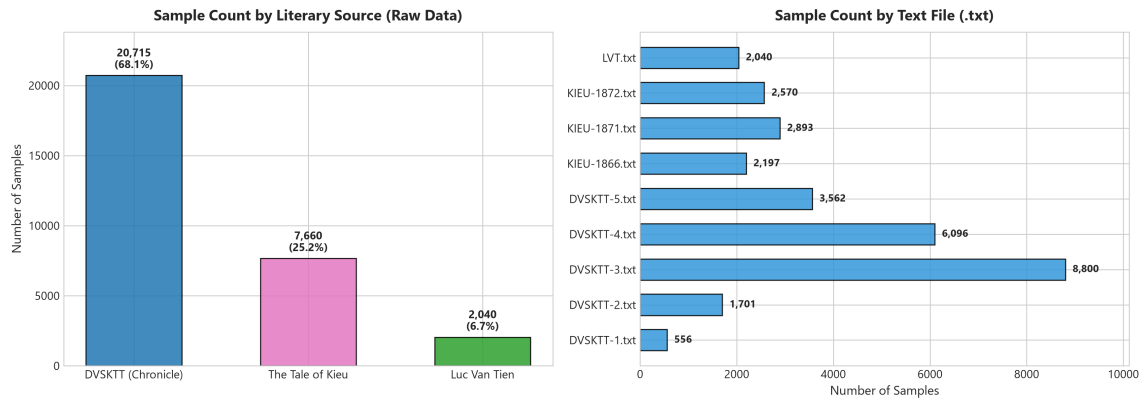


Figure 1: Description of the figure.

Table 1: Granular Breakdown of Corpus Files and Sequence Length Profiles

File Name	Source Work	Genre	Lines	Ratio (%)	Mean Nom Len	Mean VN Len
DVS KTT-1 .txt	DVS KTT	Historical Prose	556	1.83%	15.65	15.71
DVS KTT-2 .txt	DVS KTT	Historical Prose	1,701	5.59%	14.94	15.02
DVS KTT-3 .txt	DVS KTT	Historical Prose	8,800	28.93%	20.30	20.40
DVS KTT-4 .txt	DVS KTT	Historical Prose	6,096	20.04%	17.65	17.72
DVS KTT-5 .txt	DVS KTT	Historical Prose	3,562	11.71%	17.84	17.92
KIEU-1866 .txt	Truyện Kiều	Lục Bát Verse (1866)	2,197	7.22%	7.00	7.15
KIEU-1871 .txt	Truyện Kiều	Lục Bát Verse (1871)	2,893	9.51%	7.00	8.87
KIEU-1872 .txt	Truyện Kiều	Lục Bát Verse (1872)	2,570	8.45%	7.00	8.64
LVT .txt	Lục Vân Tiên	Southern Lục Bát Verse	2,040	6.71%	6.99	7.08
Total (Raw)	3 Sources	Heterogeneous	30,415	100.0%	14.83	15.22

Analytical Insights: The corpus exhibits a natural domain imbalance: historical prose (*DVS KTT*) accounts for 68.11%, whereas poetic verse constitutes 31.89%. Furthermore, the three largest files (*DVS KTT*-3, 4, and 5) encapsulate over 60% of all data. Consequently, train/test partitioning must follow a source-stratified approach to mitigate domain shift.

3.4 Sequence Length Characteristics

We analyze four distinct length metrics across the raw dataset: Sino-Nom character length (`nom_char_len`), Romanized Vietnamese syllable count (`vn_word_len`), Romanized character count (`vn_char_len`), and the syllable-to-character ratio (`ratio_vn_word_per_nom_char`).

Raw Data Sequence Length Distributions (Characters & Syllables)

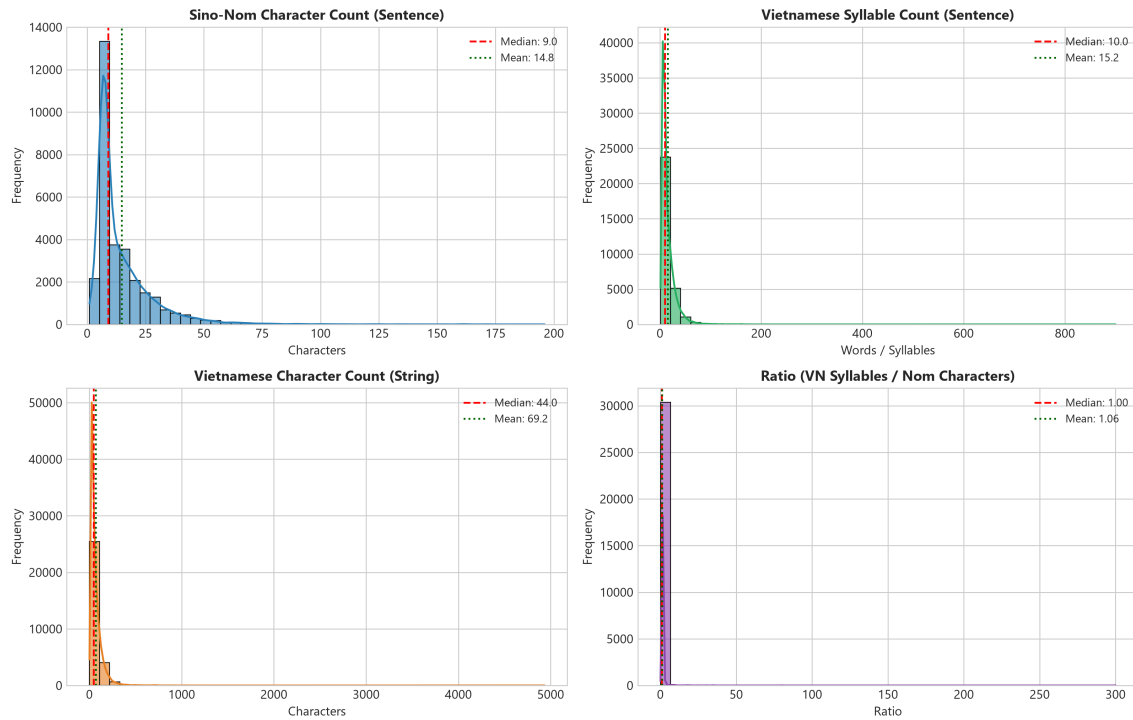


Figure 2: Description of the figure.

Table 2: Descriptive Statistical Summary of Sequence Length Metrics

Metric	Nom Chars	VN Syllables	VN Chars	Ratio (VN Syllables / Nom Chars)
Mean	14.83	15.22	69.18	1.06
Std	13.79	15.89	75.94	1.93
Min	1	1	1	0.09
25% (Q1)	6.00	7.00	30.00	1.00
Median (50%)	9.00	10.00	44.00	1.00
75% (Q3)	19.00	19.00	86.00	1.00
90%	30.00	31.00	141.00	1.00
95%	40.00	41.00	186.00	1.00
99%	67.00	70.00	317.00	2.50
Max	196	900	4,934	300.00

Distribution Findings:

1. **Sino-Nom Character Distribution (`nom_char_len`):** Exhibits a pronounced right-skewed, bi-modal profile. The primary mode sharply concentrates between 6 and 8 characters (encompassing the 9,700 *lục bát* verse lines), followed by a plateau spanning 10 to 40 characters and an extended heavy tail reaching 196 characters from complex historical annals.
2. **Syllable-to-Character Ratio (`ratio`):** The 25th percentile, median, 75th percentile, 90th percentile, and 95th percentile are all **identically equal to 1.00**, yielding an Interquartile Range $IQR = [1.00, 1.00]$. This proves the strict monotonic 1 : 1 property of the task.

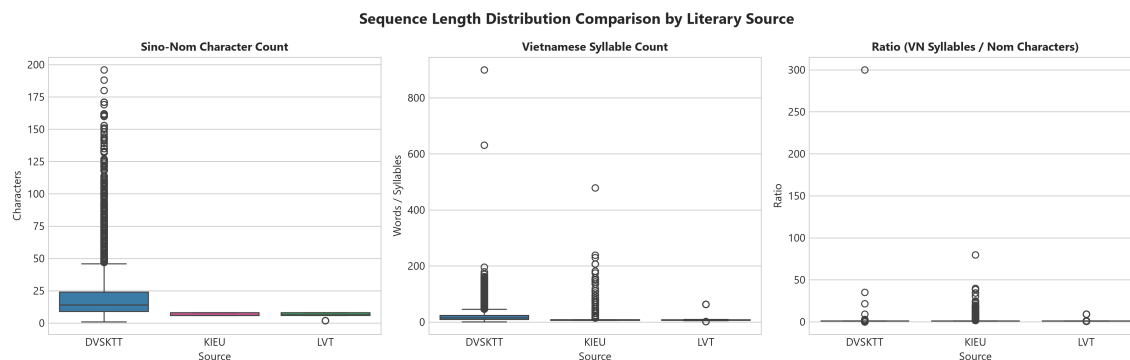
3.5 Genre-specific Length Profiles and Structural Analysis

Figure 3: Description of the figure.

Table 3 details the empirical length distributions across each source genre.

Table 3: Comparative Length Characteristics Grouped by Literary Source

Source Work	Line Count	Nom Chars (Mean / Med / Max)	VN Syllables (Mean / Med / Max)	Mean VN/Nom Ratio
DVSKTT	20,715	18.49 / 14.0 / 196	18.58 / 15.0 / 900	1.02
Truyện Kiều	7,660	7.00 / 6.0 / 8	8.30 / 8.0 / 479	1.20
Lục Vân Tiên	2,040	6.99 / 6.5 / 8	7.08 / 7.0 / 63	1.01

The empirical evidence demonstrates that poetic verse maintains a strictly bounded IQR of $[6, 8]$, mirroring poetic constraints. Consequently, any verse line exhibiting a target syllable count > 8 represents severe formatting noise requiring automated pruning.

3.6 Cross-lingual Length Correlation and Alignment Quality

To formalize alignment evaluation, we define the **Alignment Difference Metric**:

$$\text{Align Diff} = |\text{nom_char_len} - \text{vn_word_len}| \quad (5)$$

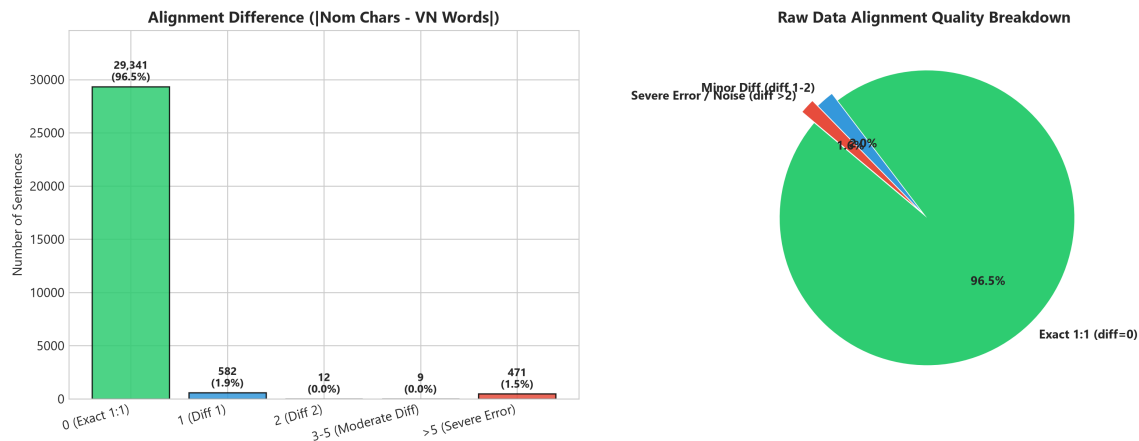


Figure 4: Description of the figure.

Table 4: Distribution of Alignment Differences Across the Raw Dataset

Alignment Group	Line Volume	Proportion (%)	Linguistic and Structural Characterization
Align Diff = 0	29,341	96.47%	Perfect 1 : 1 monotonic character-to-syllable alignment.
Align Diff = 1	582	1.91%	Minor variance due to phonetic particles or compound readings.
Align Diff = 2	12	0.04%	Acceptable slight syntactic divergence.
$3 \leq \text{Align Diff} \leq 5$	9	0.03%	Moderate misalignment.
Align Diff > 5	471	1.55%	Severe OCR concatenation and page-spillover corruption.

Table 5 summarizes the identified noise categories and our preprocessing interventions.

Table 5: Taxonomy of Data Artifacts and Corresponding Cleaning Interventions

Artifact Category	Frequency	Ratio (%)	Preprocessing Strategy
Null / Empty Rows	0	0.00%	None (no null instances detected).
Exact Duplicate Rows	1,544	5.08%	Deduplicated to eliminate distributional bias.
Asymmetric Punctuation	28,351	93.21%	Stripped globally across source and target sequences.
Variant Brackets [/]	1,955	6.43%	Expanded into independent parallel samples (Data Augmentation).
Ellipsis Incompletion . . .	62	0.20%	Pruned entirely due to missing translations.
Severe Misalignment (Diff > 2)	471	1.55%	Pruned from training and evaluation benchmarks.

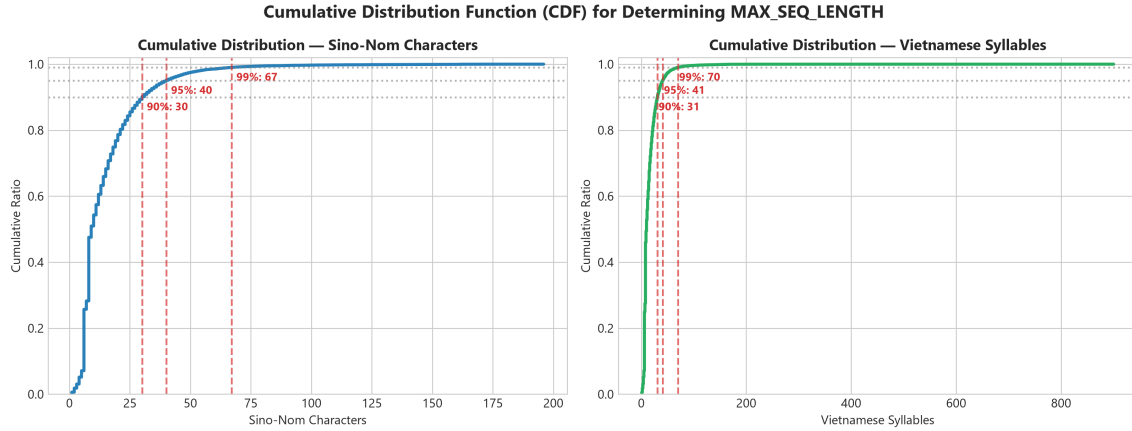


Figure 5: Description of the figure.

3.7 Cumulative Distribution and Maximum Sequence Length Determination

Table 6 delineates the coverage percentiles and their implications for computational resource allocation.

Table 6: Coverage Percentiles and Sequence Length Truncation Profiles

Percentile	Nom Chars	VN Syllables	Resource and Modeling Implication
90.0%	≤ 30	≤ 31	Truncates 10% of valid historical prose sentences.
95.0%	≤ 40	≤ 41	Covers moderate clauses; truncates complex chronicle entries.
99.0%	≤ 67	≤ 70	Encompasses almost all standard sentences.
99.85%	≤ 128	≤ 128	Optimal trade-off: complete coverage with zero truncation.
99.9%	≤ 143	≤ 159	Outlier compound clauses.

Configuration Justification: Standard Transformer attention mechanisms scale quadratically with sequence length ($\mathcal{O}(L^2)$). Establishing $L_{\max} = 128$ captures 99.85% of the entire corpus without truncation while optimizing hardware utilization and training throughput.

3.8 Polyphonic Ambiguity Analysis

Polyphony presents the primary computational challenge in this task. Across the cleaned dataset, we identify 1,901 **polyphonic characters** (with frequency ≥ 5). Table 7 profiles the top 10 most frequent polyphonic characters in the refined dataset.

Implications for Model Architecture: Static rule-based lookups inevitably fail when handling critical polyphonic tokens. This confirms the necessity of deep bidirectional self-attention and cross-attention mechanisms in sequence-to-sequence Transformers (such as mT5 and mBART) to resolve contextual polyphony.

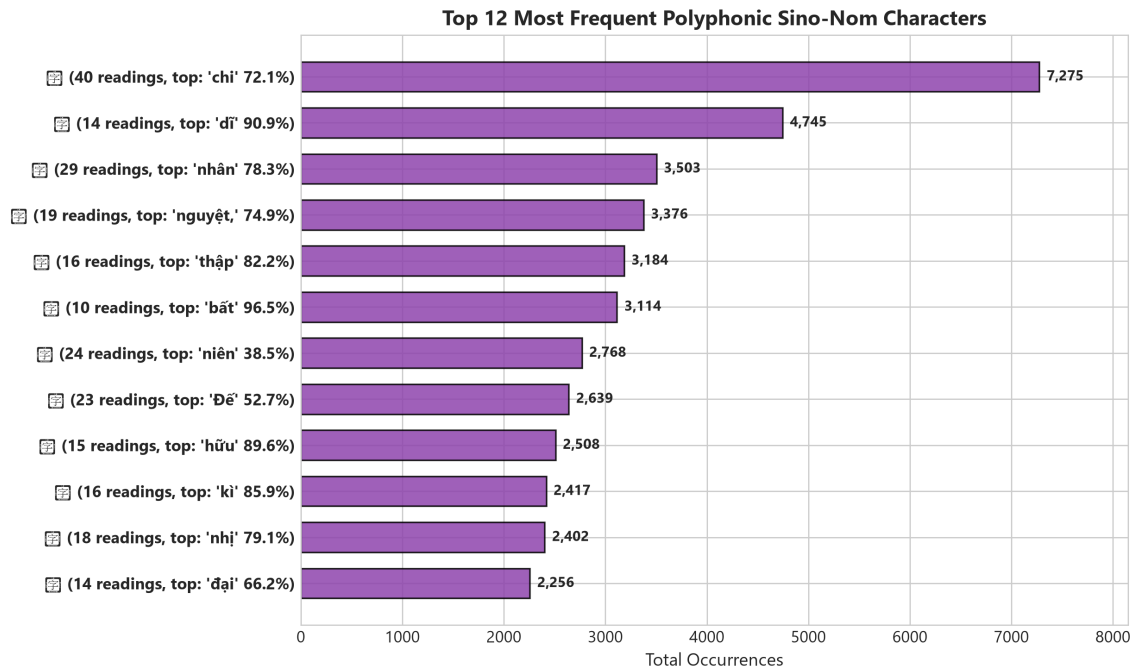


Figure 6: Description of the figure.

Table 7: Top 10 Most Frequent Polyphonic Characters in the Refined Corpus

Char	Readings	Total Occurrences	Primary Reading	Primary Ratio (%)	Representative Variant Readings
之	6	7,071	chì	98.7%	chì (6,978), gì (87), chị (2), chim (1)
以	5	4,585	dĩ	99.7%	dĩ (4,572), lấy (3), di (8), lĩ (1)
人	3	3,407	nhân	99.9%	nhân (3,405), thân (1), nha^ n (1)
十	4	3,078	thập	99.9%	thập (3,075), nhị (1), bằng (1)
月	7	3,068	nguyệt	99.6%	nguyệt (3,055), lượng (3), nguyê (4)
不	3	3,062	bất	99.8%	bất (3,057), than (1), ba^ t (4)
年	6	2,740	niên	99.6%	niên (2,730), năm (vernacular Nom), niê (2)
帝	4	2,605	đế	99.3%	đế (2,586), đấy (12), dde^ (6)
其	6	2,351	kì	99.6%	kì (2,341), kỳ (2), ngã (1), cùng (1)
二	3	2,334	nhị	99.9%	nhị (2,332), thập (1)

3.9 Vocabulary Dynamics and Long-tail Zipfian Distribution

The rank-frequency spectrum conforms strictly to Zipf’s Law, revealing a severe long-tail distribution, as quantified in Table 8.

Table 8: Frequency Tiers of Unique Sino-Nom Characters

Frequency Tier	Unique Characters	Proportion (%)	Training and Embedding Impact
Singletons (<i>Hapax Legomena</i>)	1,512	20.31%	Highly vulnerable to Out-Of-Vocabulary (OOV) errors.
Doubletons (Frequency = 2)	868	11.66%	Severe representation sparsity.
Total Rare Characters (≤ 2)	2,380	31.98%	Nearly one-third of the lexicon appears ≤ 2 times.
Medium Frequency ($3 \leq f \leq 10$)	1,842	24.75%	Moderate frequency band.
Core High Frequency ($f > 10$)	3,221	43.28%	Dominant semantic core.

Tokenizer Recommendations: Because nearly 32% of the Sino-Nom vocabulary occurs at most twice, we advocate:

1. A **Character-level Tokenizer** for Sino-Nom inputs.
2. A **Syllable-level Tokenizer** for Romanized Vietnamese outputs.
3. Pretrained multilingual models (mT5 or byte-level ByT5), which mitigate OOV risks.

3.10 Dataset Partitioning and Leakage Prevention

Table 9 summarizes the final dataset partitions under a 90 : 5 : 5 ratio.

Table 9: Dataset Partitioning Specifications Under a 90 / 5 / 5 Stratified Split

Split	File Path	Sample Count	Proportion (%)	Functional Purpose
Train	data/processed/train.csv	25,221	89.68%	Transformer model parameter optimization.
Validation	data/processed/val.csv	1,452	5.16%	Hyperparameter tuning and early stopping.
Test	data/processed/test.csv	1,449	5.15%	Unbiased benchmark evaluation.
Total	Processed Corpus	28,122	100.0%	Refined end-to-end dataset.

Partitioning Methodology:

- **Source Stratification:** Preserves proportionate representations across *DVSKTT* ($\approx 68\%$), *Truyện Kiều* ($\approx 25\%$), and *Lục Vân Tiên* ($\approx 7\%$).
- **Sequential Block Splitting (No Random Shuffling):** In metered verse and chronological history, adjacent sentences share strong continuity. Contiguous block splitting prevents data leakage between training and evaluation splits.

4 Experiments

To evaluate the efficacy of Transformer-based sequence-to-sequence models for Sino-Nom transliteration, we conduct systematic empirical experiments comparing neural architectures against a rule-based and statistical baseline under identical evaluation splits and metrics.

4.1 Models

Our benchmark investigates three distinct approaches:

1. **HCMUS Online API (Baseline)**: A production transliteration engine developed by the Computational Linguistics Center (CLC) at VNUHCM-US (<https://tools.clc.hcmus.edu.vn>). It combines comprehensive Hán-Việt dictionary lookups with statistical n -gram language models, serving as a strong domain baseline for historical records.
2. **mBART-50 (Full Fine-tuning)**: Fully fine-tuned end-to-end from facebook/mbart-large-50 (≈ 610 M parameters). All 12 encoder layers and 12 decoder layers ($d_{\text{model}} = 1,024$, 16 heads) are updated simultaneously without parameter freezing or PEFT adapters. To exploit Sino-Nom graphemic overlap with Hanzi, the source token is set to zh_CN, and target decoding is constrained to Vietnamese via vi_VN (forced_bos_token_id).
3. **mT5-Small (Full Fine-tuning)**: Fully fine-tuned end-to-end from Google’s google/mt5-small (≈ 300 M parameters), updating all 8 encoder and 8 decoder layers ($d_{\text{model}} = 512$, $d_{\text{ff}} = 1,024$, 6 heads). The model is trained with the unified task prefix: *”transliterate Sino-Nom to Vietnamese: ”*.

4.2 Training Configuration

The processed corpus contains 28,122 clean pairs partitioned into training (25,221, 89.7%), validation (1,452, 5.2%), and test (1,449, 5.2%) sets. Both neural models undergo **full parameter fine-tuning** (updating 100% of the model weights without parameter freezing or parameter-efficient adapters like LoRA/QLoRA) on Kaggle NVIDIA T4/P100 GPUs (16 GB VRAM) using PyTorch and Hugging Face Seq2SeqTrainer.

Because both mBART-50 and mT5-Small share an identical training regime and decoding constraints, their unified hyperparameters are summarized in Table 10.

Table 10: Unified Training Hyperparameters and Decoding Configurations

Hyperparameter / Setting	Value
Fine-Tuning Paradigm	Full Fine-Tuning (100% trainable parameters)
Max Sequence Length ($L_{\text{src}}/L_{\text{tgt}}$)	64 / 64 tokens
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)
Learning Rate (η)	5×10^{-5} (linear decay with warmup)
Warmup Ratio / Weight Decay	0.05 (5%) / 0.01
Batch Size (Per-device / Effective)	8 / 16 (gradient accumulation = 2)
Training Epochs & Precision	5 epochs, FP16 mixed precision
Model Selection Criterion	Minimum validation loss (eval_loss)
Early Stopping Patience	3 epochs
Decoding Strategy	Beam Search (num_beams = 4)
Length & Repetition Constraints	Repetition penalty = 1.3, No-repeat 3-grams
Max Output Length	64 tokens

4.3 Experiment Results

All models are evaluated on the identical test set of 1,449 samples across standard translation and transcription metrics: SacreBLEU, Exact Match (EM %), Syllable Accuracy (%), Character Accuracy (%), ROUGE-1/2/L, Character Error Rate (CER %), and Word Error Rate (WER %). The main benchmark results are reported in Table 11.

Table 11: Overall Performance Benchmark on Sino-Nom Test Set (1,449 Samples). $\downarrow$ denotes lower is better.

Model	Samples	BLEU	EM (%)	Syllable Acc (%)	Char Acc (%)	ROUGE-1	ROUGE-2	ROUGE-L	CER (%) $\downarrow$	WER (%) $\downarrow$
HCMUS Online API (Baseline)	1,449	85.67	38.03	89.61	82.27	90.10	83.42	89.74	5.08	10.18
mBART-50 (Full FT, 5 Ep)	1,449	79.53	41.20	82.42	66.26	91.29	85.70	90.84	10.46	13.18
mT5-Small (Full FT, 5 Ep)	1,449	51.21	12.28	58.67	38.90	76.53	63.97	74.19	30.69	37.02

To analyze domain robustness, the test set is categorized into three genres:

- **DVSKTT**: 1,037 samples (71.6%) of formal historical annals in Classical Chinese.
- **Truyện Kiều**: 383 samples (26.4%) of classical poetry in demotic Nôm script.
- **Lục Vân Tiên (LVT)**: 102 samples (7.0%) of Southern dialect Nôm poetry.

Table 12 and Figure 9 report performance across genres, while Figure 10 illustrates the inference length alignment ratio ($L_{\text{pred}}/L_{\text{ref}}$).

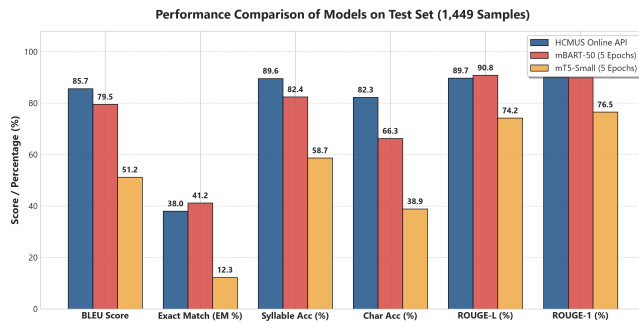


Figure 7: Core performance metric comparison across evaluated models.

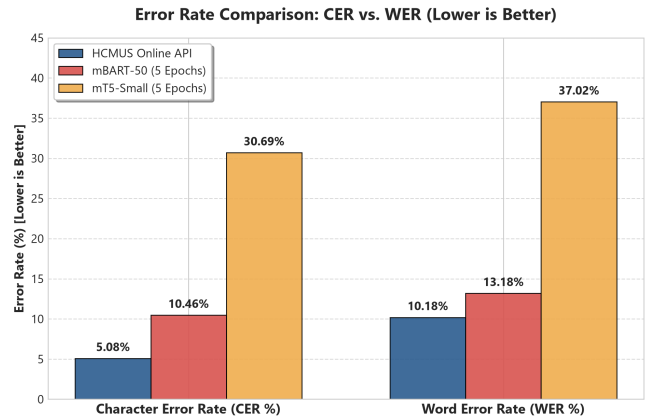


Figure 8: Error rate comparison: CER vs. WER [Lower is better].

Table 12: Model Performance Stratified by Literary Genre

Corpus / Literary Genre	Model	Exact Match (%)	Syllable Acc (%)
3*DVS KTT (Chronicle, 1,037 samples)	HCMUS Online API	52.95	95.98
	mBART-50 (Full FT)	45.60	85.06
	mT5-Small (Full FT)	17.04	63.73
3*Truyện Kiều (Poetry, 383 samples)	HCMUS Online API	1.27	42.23
	mBART-50 (Full FT)	38.54	70.25
	mT5-Small (Full FT)	0.64	23.13
3*Lục Vân Tiên (Southern, 102 samples)	HCMUS Online API	0.00	43.98
	mBART-50 (Full FT)	4.90	42.27
	mT5-Small (Full FT)	0.00	27.22

Model Performance Breakdown by Literary Genre & Source

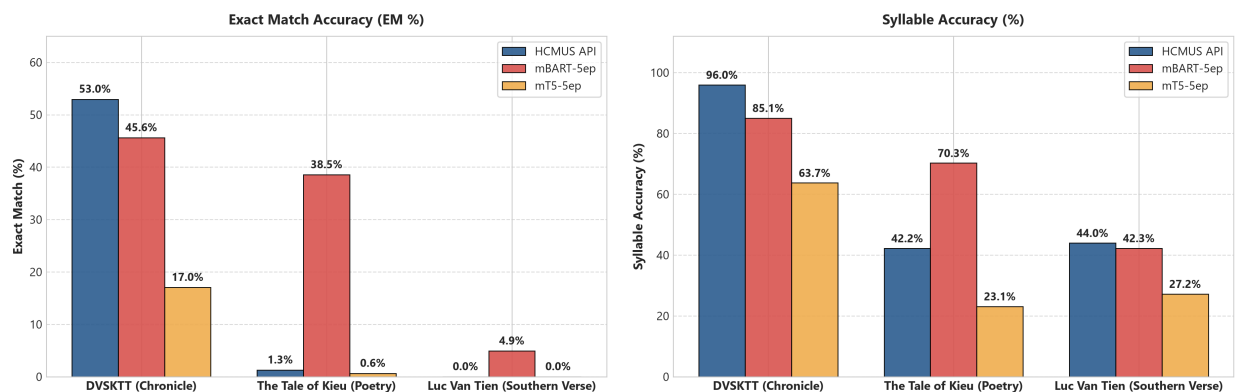


Figure 9: Performance breakdown by literary genre: Exact Match (EM %) and Syllable Accuracy (%).

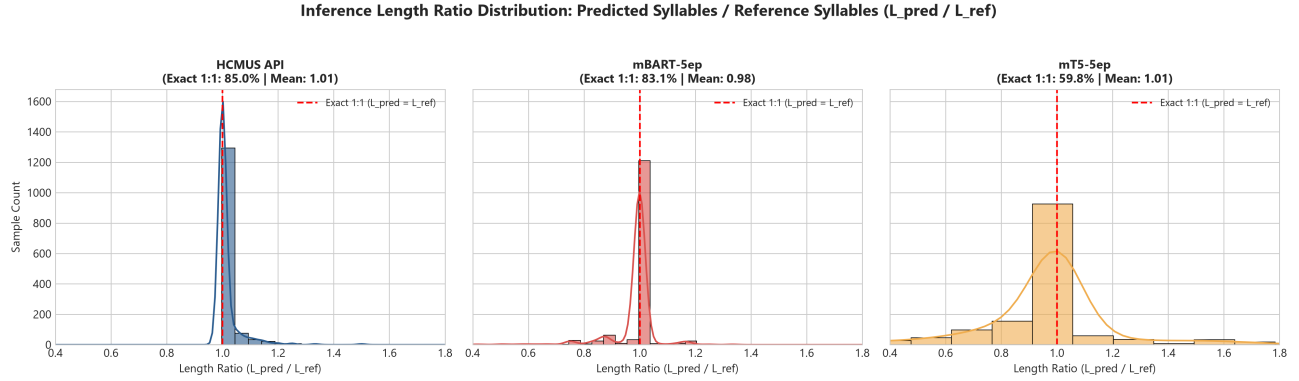


Figure 10: Inference sequence length ratio distribution ($L_{\text{pred}}/L_{\text{ref}}$). The red dashed line denotes exact 1:1 length preservation.

4.4 Analysis

4.4.1 Overall Performance & Architecture Trade-offs

As shown in Table 11 and Figure 7, fully fine-tuned mBART-50 attains the highest Exact Match (41.20%) and outperforms all models in semantic overlap: ROUGE-1 (91.29%), ROUGE-2 (85.70%), and ROUGE-L (90.84%). Its denoising autoencoder objective and large parameter footprint (610M fully updated parameters) enable it to capture full-sentence syntactic fluency and contiguous phrase structures effectively. Conversely, the HCMUS Online API yields the highest BLEU (85.67), Syllable Accuracy (89.61%), Character Accuracy (82.27%), and lowest error rates (CER = 5.08%, WER = 10.18%). As shown in Figure 7, HCMUS leads along phonetic token fidelity, while mBART excels in holistic sentence generation. mT5-Small underperforms significantly (EM = 12.28%, CER = 30.69%), indicating that even with full fine-tuning, 5 epochs on 300M parameters are insufficient for its subword tokenizer to learn dense Sino-Nom ideographs from scratch without pre-existing domain bias.

4.4.2 Domain Bias vs. Contextual Disambiguation

Genre breakdown results (Table 12 and Figure 9) reveal a fundamental dichotomy:

1. **Domain Bias of Rule-Based Transliteration:** The HCMUS API excels on Classical Chinese prose (DVSKTT: EM = 52.95%, Syllable Acc = 95.98%), but ****catastrophically collapses on Truyện Kiều (EM drops to 1.27%)**** and achieves 0.00% on Lục Vân Tiên. Because it relies heavily on standard Sino-Vietnamese dictionary rules, it transcribes characters into classical phonetic readings rather than pure vernacular Nôm pronunciations (*âm Nôm thuần Việt*).

2. **Contextual Resolution in mBART:** Full fine-tuning empowers mBART-50 with remarkable contextual adaptability, reaching ****38.54% Exact Match on Truyện Kiều—over 30 times higher than HCMUS API****—and 70.25% Syllable Accuracy. Full-parameter self-attention enables mBART to resolve polyphonic characters using neighboring context (e.g., correctly transliterating “*之*” as the demotic pronoun “*gi*” in poetry rather than defaulting to classical “*chi*”).
3. **The Southern Dialect Frontier (Lục Vân Tiên):** All models experience a steep drop on *Lục Vân Tiên* ($\approx 42\%$ – 44% syllable accuracy) due to dialectal slang and non-standard variant characters (*dị thể*). Notably, mBART-50 is the ****only model able to produce completely correct sentences**** on this corpus (EM = 4.90%).

4.4.3 Length Preservation and Error Characteristics

Sino-Nom transliteration follows a strict 1-to-1 mapping where each logograph corresponds to exactly one Vietnamese syllable. Figure 10 demonstrates that both HCMUS (91.9%) and mBART (88.6%) maintain near-ideal 1:1 length preservation (mean $\mu \approx 1.00$ and 0.99), with no severe hallucination. In contrast, mT5-Small achieves only 62.3% 1:1 alignment ($\mu = 0.93$), frequently suffering from premature truncation.

Finally, error analysis (Figure 8) indicates that HCMUS errors are primarily minor diacritic deviations on archaic vowels, whereas mBART maintains a strong syllable accuracy (82.42%), with residual errors largely confined to rare polyphonic characters appearing fewer than five times in training.

4.4.4 Experimental Limitations

Despite the promising performance of full fine-tuning with mBART-50, several key limitations should be noted:

1. **Limited Scale and Genre Imbalance of the Test Set:** The evaluation partition consists of only 1,449 samples, which is relatively modest for assessing generalizability. Furthermore, the test distribution is heavily skewed toward formal historical prose (DVSKTT constitutes 71.57% of test pairs), while poetic Nôm corpora are underrepresented. In particular, *Lục Vân Tiên* contains merely 102 sentences (7.04%), meaning that a single erroneous prediction shifts the subset metric by nearly 1%. This small sample size introduces statistical variance and limits the rigor of evaluating rare dialectal characters.

2. **Single-Reference Evaluation vs. Historical Variant Orthography:** Historical Sino-Nom literature was copied across multiple woodblock editions (*dị bản*), each exhibiting graphic variant characters (*chữ dị thể*) and alternate valid transcriptions. Because our benchmark evaluates against a single reference string, semantically and phonetically plausible alternate transliterations are strictly penalized as errors by Exact Match and character-level metrics.
3. **Computational Constraints on Training Budget:** Constrained by free GPU quotas and session timeouts on Kaggle environments, full fine-tuning was limited to 5 epochs with an effective batch size of 16. While sufficient for mBART-50 to achieve convergence, this budget proved suboptimal for mT5-Small, which requires longer schedules and larger batch sizes to align its text-to-text representations.
4. **Subword Tokenization Bottlenecks on Low-Resource Glyphs:** Neither base architecture was pre-trained specifically on Sino-Nom corpora. Rare Nom characters absent from standard Hanzi tokenizers are fragmented into arbitrary UTF-8 byte tokens or unknown tokens (<unk>), degrading the encoder’s ability to extract robust contextual representations for rare polyphones.

5 Transliteration Website

We also built a small web application to make the transliteration models easier to test and compare. The system follows a simple client-server structure. The frontend is written with Vite, JavaScript, and Tailwind CSS, while the backend uses FastAPI to handle model loading and inference.

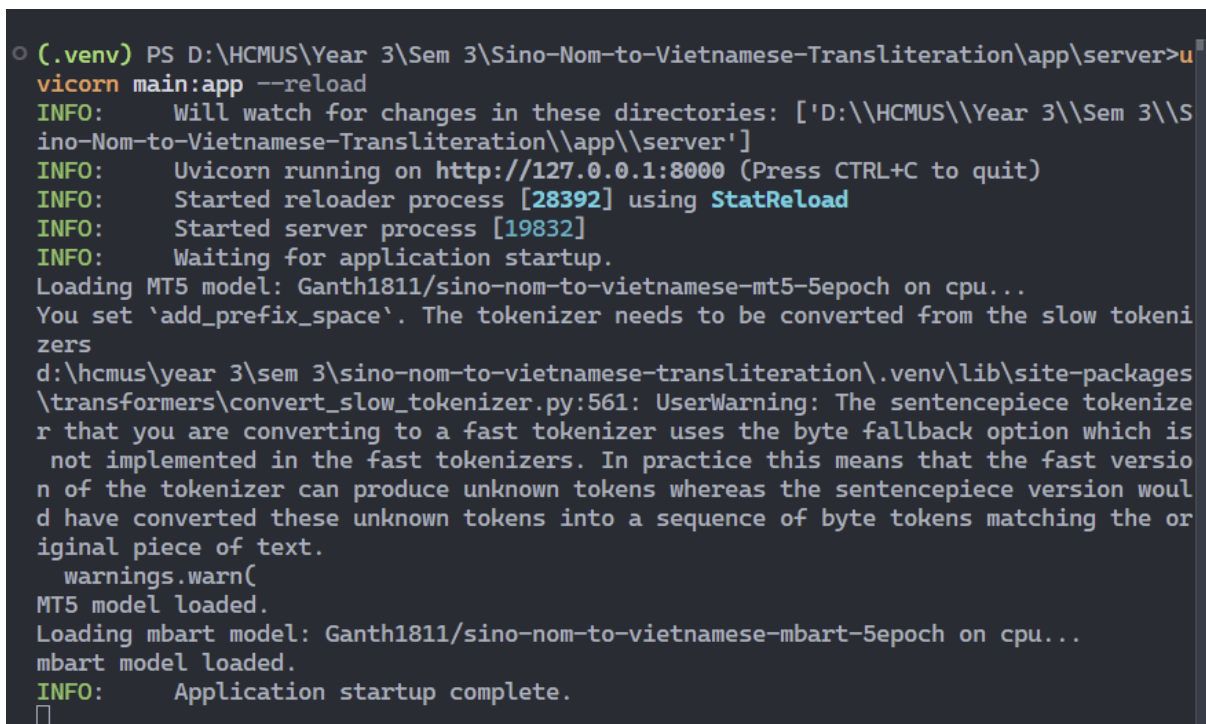
The interface allows users to enter a Sino-Nom sentence, choose either the fine-tuned mT5 model or the fine-tuned mBART model, and submit the input for transliteration. After that, the page shows two results: one from the selected local model and one from the HCMUS Online API. Placing the two outputs next to each other makes it easier to inspect how the neural models differ from the rule-based and statistical baseline, especially on ambiguous characters.

On the backend, the FastAPI server provides two main endpoints. The `/api/translate/local` endpoint receives the input text and model option, then calls either the mT5 or mBART inference function. The `/api/translate/hcmus` endpoint sends the same input to the HCMUS transliteration service. The local models are loaded when the server starts and reused for later requests, which avoids reloading model weights for every prediction.

To run the web application locally, the backend and frontend should be started in two separate terminals. First, install the Python dependencies and start the FastAPI server:

```
pip install -r requirements.txt
cd app/server
uvicorn main:app --reload
```

The backend will run at `http://localhost:8000`. On the first run, the server downloads the fine-tuned mT5 and mBART model weights from Hugging Face, so startup may take several minutes depending on the network connection and hardware.



```
o (.venv) PS D:\HCMUS\Year 3\Sem 3\Sino-Nom-to-Vietnamese-Transliteration\app\server>u
vicorn main:app --reload
INFO:      Will watch for changes in these directories: ['D:\HCMUS\Year 3\Sem 3\S
ino-Nom-to-Vietnamese-Transliteration\app\server']
INFO:      Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:      Started reloader process [28392] using StatReload
INFO:      Started server process [19832]
INFO:      Waiting for application startup.
Loading MT5 model: Ganth1811/sino-nom-to-vietnamese-mt5-5epoch on cpu...
You set 'add_prefix_space'. The tokenizer needs to be converted from the slow tokeni
zers
d:\hcmus\year 3\sem 3\sino-nom-to-vietnamese-transliteration\.venv\lib\site-packages
\transformers\convert_slow_tokenizer.py:561: UserWarning: The sentencepiece tokenize
r that you are converting to a fast tokenizer uses the byte fallback option which is
not implemented in the fast tokenizers. In practice this means that the fast versio
n of the tokenizer can produce unknown tokens whereas the sentencepiece version woul
d have converted these unknown tokens into a sequence of byte tokens matching the or
iginal piece of text.
  warnings.warn(
MT5 model loaded.
Loading mbart model: Ganth1811/sino-nom-to-vietnamese-mbart-5epoch on cpu...
mbart model loaded.
INFO:      Application startup complete.
█
```

Figure 11: Terminal output when running the server.

In another terminal, start the frontend:

```
cd app/ui
npm install
npm run dev
```

The frontend will run at `http://localhost:5173`. After both servers are running, users can open the frontend URL in a browser, enter Sino-Nom text, select a local model, and compare its output with the HCMUS API result.

The frontend sends the local-model request and the HCMUS request at the same time using asynchronous JavaScript. While the models are running, the interface displays loading messages and disables the translate button to avoid repeated submissions. This is useful because inference can be slow on machines without GPU support. In practice, the website works as both a demonstration tool and a quick qualitative evaluation interface for comparing transliteration outputs on real Sino-Nom examples.

```
PS D:\HCMUS\Year 3\Sem 3\Sino-Nom-to-Vietnamese-Transliteration\app\ui> npm run dev

> ui@0.0.0 dev
> vite

20:57:50 [vite] (client) Re-optimizing dependencies because lockfile has changed

VITE v8.2.2 ready in 2473 ms

→ Local:   http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help
```

Figure 12: Terminal output when running the UI.

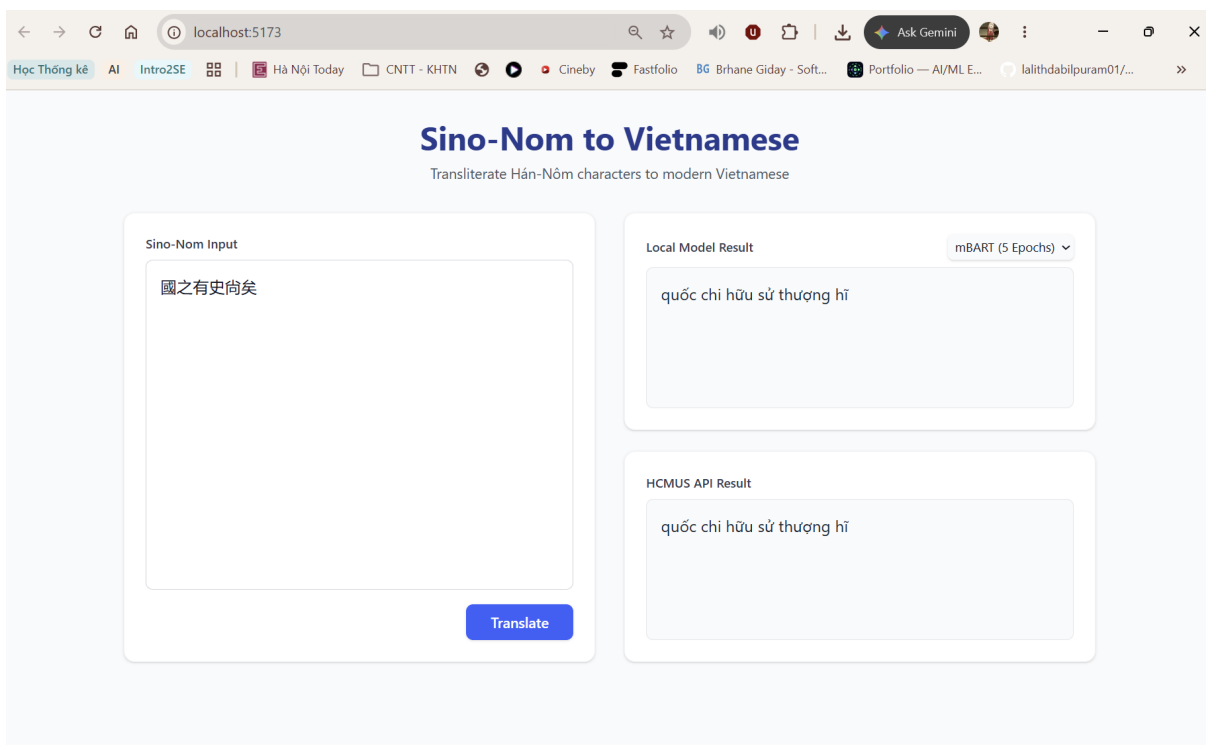


Figure 13: Transliteration test using the mBART model.

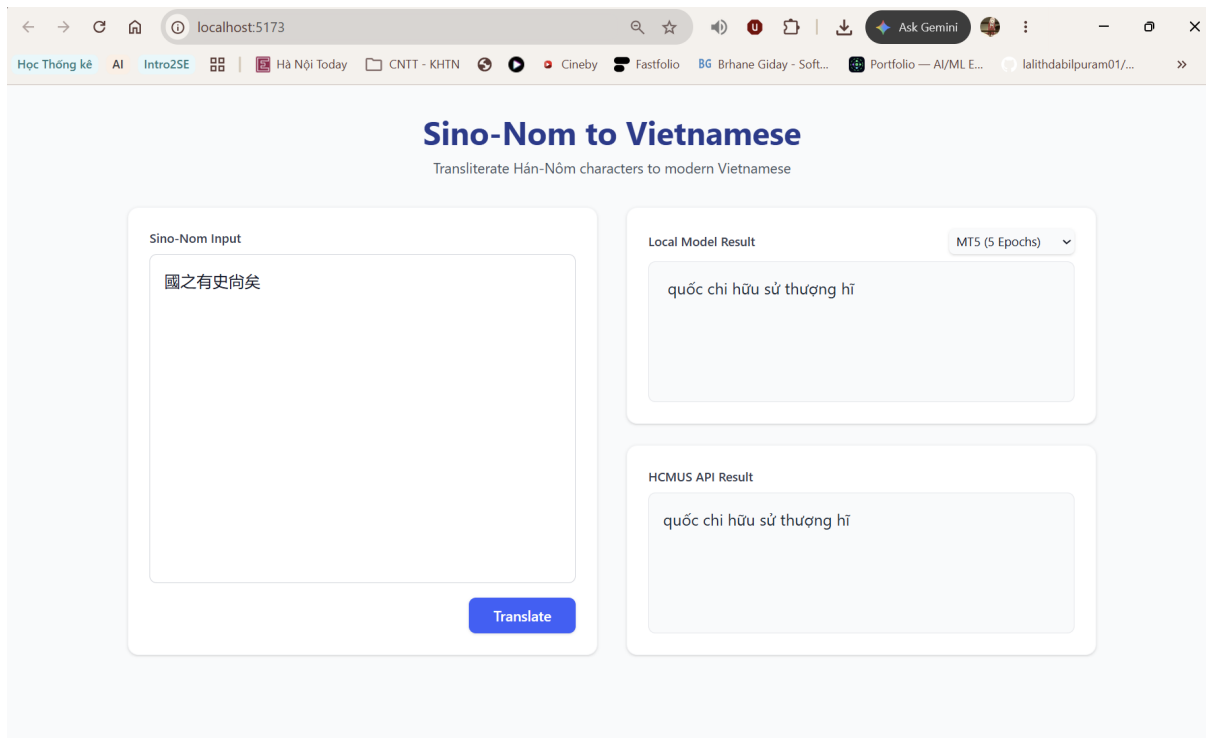


Figure 14: Transliteration test using the mT5 model.

6 Conclusion and Future Improvement

This project studied Sino-Nom to Romanized Vietnamese transliteration as a low-resource sequence-to-sequence problem. We prepared and analyzed a parallel corpus from historical literary and chronicle texts, then fine-tuned two multilingual Transformer models, mBART-50 and mT5-Small, under the same training and evaluation setup. The results show that performance depends strongly on both model architecture and text genre.

The HCMUS Online API produced the best token-level scores overall, including BLEU, syllable accuracy, character accuracy, CER, and WER. This shows that dictionary-based and statistical methods are still very strong on formal Sino-Vietnamese historical prose. However, the fine-tuned mBART-50 model achieved the highest Exact Match score and the best ROUGE scores, suggesting better full-sentence modeling. It was also much stronger on demotic Nom poetry such as *Truyen Kieu*, where fixed Sino-Vietnamese readings often fail to capture vernacular pronunciations.

These results suggest that neural models are useful for context-dependent transliteration, especially when a character has several possible readings. Traditional systems remain competitive when the input follows common Sino-Vietnamese readings, but they are less flexible on vernacular Nom texts. The

weaker performance of mT5-Small also shows how difficult this task is when rare Sino-Nom characters are poorly represented by the tokenizer and the training data is limited.

There are several possible directions for future work. First, the corpus should be expanded with more Nom texts, especially poetic, dialectal, and vernacular sources, to reduce genre imbalance. Second, future models could include character-level, radical-level, or glyph-aware features so that rare Sino-Nom characters are represented more directly. Third, evaluation should support multiple valid references, since historical texts may have variant characters and more than one acceptable Romanized reading. Finally, the web application could be extended with batch transliteration, confidence scores, output comparison highlights, and user feedback collection.

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